applied sciences

MDPI

Article

Functional Motor Profiling and Performance Correlates in Youth Football Players: A Neuroscience Perspective

Vladimir Potop [1,2,3,*], Bogdan Andrei Pana [4], Liviu Emanuel Mihailescu [3], Ilie Mihai [3], Paul Florinel Visan [3], Niculina Liliana Mihailescu [1], Constantin Ciorba [5], Dumitru Barbu [6] and Ion Mihaila [1,3]

[1] Doctoral School of Science of Sports and Physical Education, National University of Science and Technology POLITEHNICA Bucharest, University Centre Pitesti, 110040 Pitesti, Romania; niculina.mihailescu@upb.ro (N.L.M.); ion.mihaila@upb.ro (I.M.)
[2] Institute of Physical Education and Sport, Moldova State University, MD-2024 Chisinau, Moldova
[3] Department of Physical Education and Sport, National University of Science and Technology POLITEHNICA Bucharest, University Centre Pitesti, 110040 Pitesti, Romania; liviu.mihailescu74@upb.ro (L.E.M.); ilie.mihai@upb.ro (I.M.); paul_florinel.visan@upb.ro (P.F.V.)
[4] Sports Club Otopeni, 075100 Otopeni, Romania; sega_andrey@yahoo.com
[5] Ion Creangă' Pedagogical State University, MD-2069 Chisinau, Moldova; ciorba.constantin@upsc.md
[6] University of Craiova, 200585 Craiova, Romania; dumitru.barbu@edu.ucv.ro
* Correspondence: vladimir.potop@upb.ro; Tel.: +40-721-324-867

Featured Application

The findings of the present study may support football coaches and practitioners in optimizing psychomotor and neuromuscular training strategies during the developmental stages of youth football players. The proposed multidimensional functional profiling model may facilitate individualized training management, improve movement efficiency, and support football-specific performance optimization during the competitive season.

Abstract

Psychomotor development plays an important role in neuromuscular regulation, movement efficiency, and football-specific performance during early adolescence. This study investigated changes associated with an integrated psychomotor-oriented intervention program in psychomotor abilities, technical performance, and functional physiological responses in U13 football players. A longitudinal quasi-experimental pretest–posttest design was applied to 17 male youth football players during the competition season. The 12-week intervention program was integrated into regular football training and included psychomotor, neuromuscular, coordinative, and football-specific exercises. Psychomotor assessment included sprint performance, postural stability and control, agility, and reactive neuromuscular parameters, complemented by football-specific technical tests and functional match monitoring using the Polar Vantage V system. An external comparative U13 team (n = 17) was additionally evaluated for football-specific technical performance. Significant within-group improvements were observed in 20-m sprint performance (Δ = −0.19 s; $p < 0.001$; d = −1.13), lateral bipedal postural stability (Δ = +20%; $p < 0.001$; d = 1.35), and vertical bipedal postural control (Δ = +10%; $p = 0.006$; d = 0.77). All three technical tests improved significantly within the intervention group (Test 1: $p = 0.002$, d = 0.88; Test 2: $p = 0.016$, d = 0.75; Test 3: $p = 0.037$, d = 0.63), while between-group comparisons of change scores were not statistically significant. Functional monitoring during official match play identified a significant reduction in Speed Z3 exposure ($p = 0.016$; d = −0.70), while no statistically significant changes were observed in the other physiological or locomotor

Academic Editor: René Schwesig

Received: 1 August 2026
Revised: 3 October 2026
Accepted: 5 October 2026
Published: 12 October 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

variables. Spearman correlation analyses identified significant associations among psychomotor, technical, and functional variables ($\varrho = -0.753$ to -0.488; $p < 0.05$). Given the small sample, non-randomized design, incomplete multidimensional assessment in the comparative group, and absence of biological maturation assessment, the findings should be interpreted as preliminary evidence of within-group changes associated with psychomotor-oriented training rather than as definitive causal effects.

Keywords: psychomotor training; youth football; neuromuscular adaptation; functional profiling; motor performance; postural control; football-specific performance

1. Introduction

Youth football performance is a multidimensional process involving the interaction of physical, technical, cognitive, perceptual, and neuromotor factors. During preadolescence, these components develop concurrently and are influenced by biological growth, motor learning, training exposure, and sport-specific experience. Thus, the assessment of young football players should not be restricted to isolated measures of physical capacity or technical skill. It should also consider the functional organization of movement and the ability to regulate motor responses under changing task demands [1–6].

Psychomotor development refers to the functional integration of motor, perceptual, coordinative, and sensorimotor processes that support posture, balance, movement regulation, and goal-directed motor actions. In youth football, these processes are relevant because players must continuously integrate body position, movement speed, spatial information, and task-specific motor responses during both training and competition. During preadolescence, ongoing neural and sensorimotor development may provide an important context for the acquisition and refinement of coordinative and movement-regulation abilities [4–7].

Accordingly, psychomotor-oriented training has been associated with improvements in balance, coordination, reactive agility, sprint-related performance, and other neuromuscular characteristics in young athletes [5,8–11].

The relevance of psychomotor development may extend beyond isolated motor-test performance. Football-specific actions require the integration of postural control, locomotor speed, coordination, and technical execution, while competition activity imposes additional physiological and locomotor demands. Previous research has demonstrated beneficial effects of psychomotor and neuromuscular interventions on selected components of motor performance, including balance, sprint ability, coordination, and technical execution [12–17]. In parallel, wearable monitoring systems such as Polar technology can provide information about heart-rate responses, locomotor activity, and the intensity distribution of movement during match play [18–20]. These complementary assessment approaches may therefore provide different but interconnected perspectives on player adaptation.

However, the functional transfer of psychomotor-oriented training to football-specific performance and competitive physiological responses remains insufficiently characterized in U13 players. Available studies have frequently examined psychomotor, physical, technical, or physiological variables separately, making it difficult to determine how adaptations in one domain relate to changes in another [21–23]. Moreover, relatively few investigations have combined standardized psychomotor assessments, football-specific technical tests, and functional monitoring during competitive activity within the same

longitudinal framework [24–27]. This limitation is particularly relevant during preadolescence, when normal developmental changes and sport-specific training occur simultaneously and may influence performance results.

The present study was therefore designed around a multidimensional functional profiling approach. Psychomotor assessments were selected to characterize sprint performance, agility, postural stability and control, and vertical jumping characteristics. These represent complementary aspects of movement speed, postural control, coordination, and neuromuscular performance. Football-specific technical tests were included to examine whether changes observed in psychomotor characteristics were accompanied by changes in task-specific technical execution. In addition, functional monitoring during competition activity was used to define heart-rate and locomotor responses in an ecologically valid match environment. The combination of these measurements was intended to provide a broader description of adaptation than would be possible using a single performance domain.

An additional methodological consideration was the temporal structure of the data collection. Psychomotor and functional assessments were performed immediately before and after the 12-week integrated psychomotor intervention implemented during the return phase of the competitive season. Thus, it was possible to examine short-term changes associated with the intervention period. Football-specific technical performance was assessed at the beginning and end of the competitive season, while an external comparative U13 team from the same county championship underwent the same technical assessments. This strategy allowed the observed technical changes in the intervention group to be interpreted in relation to the evolution of technical performance in another team exposed to regular football training but not to the psychomotor-oriented intervention. The multidimensional dataset also enabled secondary analyses of relationships between psychomotor, technical, locomotor, and physiological indicators.

Based on this rationale, the primary scientific question was whether an integrated psychomotor-oriented training program incorporated into regular football preparation was associated with measurable changes in psychomotor performance. Also, we analyzed whether these changes were accompanied by differences in football-specific technical performance and functional responses during competition activity.

The aim of this study was therefore to investigate the effects of an integrated psychomotor intervention program on psychomotor abilities, technical performance, and physiological responses in U13 football players during the competition season.

We hypothesized that the intervention would be associated with improvements in postural control and sprint-related motor performance, accompanied by improvements in football-specific technical execution and changes in physiological and locomotor responses during competitive activity.

2. Materials and Methods

2.1. Study Design

This study used a longitudinal quasi-experimental repeated-measures design to examine intervention-related adaptations associated with an integrated psychomotor and neuromuscular training program in youth football players (Figure 1).

The research was conducted during the 2022–2023 competitive season within the U13 team of CS Otopeni (Romania). The study protocol included two evaluation periods corresponding to the competitive season and the intervention program.

Football-specific technical performance was assessed at the beginning of the championship season (Technical Initial Assessment—TIA) and at the end of the competitive season (Technical Final Assessment—TFA). In contrast, psychomotor and functional evaluations were conducted immediately before (PRE) and after (POST) the implementation of

the intervention program during the second phase (return round) of the championship season.

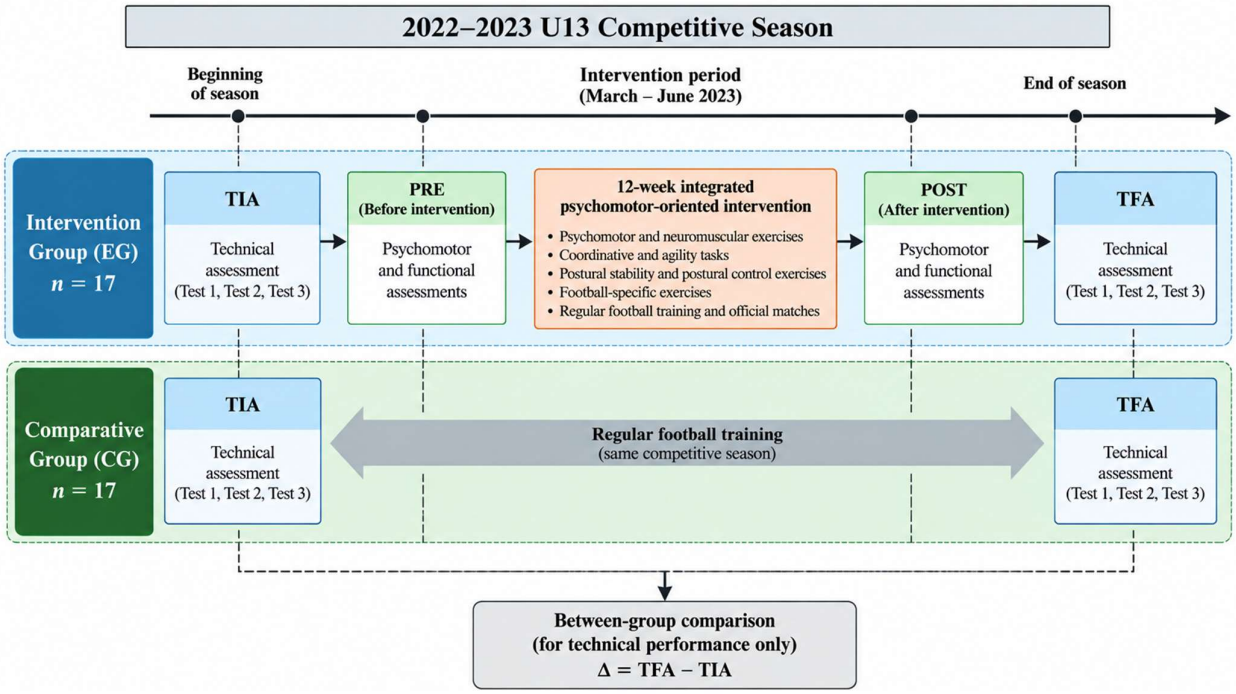


Figure 1. Longitudinal study design and assessment timeline in U13 football players. TIA, Technical initial Assessment; TFA, Technical Final Assessment; PRE, pre-intervention assessment; POST, post-intervention assessment; EG, intervention group; CG, comparative group. **Note:** Psychomotor and functional assessment were performed only in the intervention group; therefore, between-group comparisons were restricted to changes in technical performance (Δ = TFA − TIA).

The intervention program was implemented during the competitive return phase over approximately three months (March–June 2023). During this period, the players participated in regular football training sessions, official matches, and the integrated psychomotor training program.

To provide contextual information regarding the evolution of football-specific technical performance during the competitive season, an external comparative U13 football team from the same county championship was additionally evaluated. This comparative team followed its regular training program and underwent the same football-specific technical assessments (Technical Initial Assessment and Technical Final Assessment). No psychomotor intervention, physiological monitoring, or functional assessments were performed in the comparative team. Consequently, between-group analyses were restricted exclusively to football-specific technical performance.

The comparative team included players of similar chronological age and competitive level participating in the same county championship.

Psychomotor assessments were carried out in the sports hall of CS Otopeni under standardized conditions, whereas football-specific technical evaluations were performed on the training field. Functional monitoring during official matches was performed using the Polar Vantage V system combined with Polar Flow software Firmware 5.2.4 (Polar Electro Oy, Kempele, Finland). Monitoring was conducted before and after the intervention period to assess cardiovascular responses and locomotor activity across different intensity zones in real match conditions.

The multidimensional assessment framework was used to characterize relationships between psychomotor adaptation, football-specific technical performance, physiological responses, and competition performance indicators during the competitive season.

The psychomotor, neuromuscular and functional assessments were conducted within the Human Performance Research Center of the Department of Physical Education and Sport, in collaboration with the Doctoral School of Science of Sports and Physical Education, University Center Pitești, National University of Science and Technology "Politehnica" Bucharest.

2.2. Participants

The intervention group consisted of 17 male U13 football players from CS Otopeni. Additionally, an external comparative group (n = 17) was included exclusively for the assessment of football-specific technical performance. The comparative group was selected based on comparable chronological age, competitive level, training frequency, participation in the same county championship, and a similar position in the league standings during the 2022–2023 competition season. Both groups belonged to the U13 competitive category and regularly participated in organized football training and official championship matches. Baseline anthropometric characteristics indicated that the two groups were comparable. The intervention group had a mean age of 12.47 ± 0.62 years, sports experience of 5.29 ± 0.69 years, a body height of 153.24 ± 7.05 cm, a body weight of 40.97 ± 6.73 kg, and a body mass index of 17.38 ± 1.81 kg/m^2. The comparative group had a mean age of 12.41 ± 0.62 years, sports experience of 5.24 ± 0.66 years, a body height of 152.94 ± 6.90 cm, a body weight of 40.86 ± 6.83 kg, and a body mass index of 17.41 ± 2.01 kg/m^2. The comparative group underwent the same standardized technical evaluations at the beginning (Technical Initial Assessment, TIA) and at the end (Technical Final Assessment, TFA) of the competitive season but continued its regular football training without receiving the psychomotor-oriented intervention. Players in both groups regularly participated in technical–tactical and physical training sessions throughout the competition season within their respective football clubs.

Inclusion criteria were: (a) participation in all testing sessions; (b) regular participation in training sessions and official matches during the intervention period; and (c) absence of musculoskeletal injuries or medical conditions affecting performance.

Players who missed testing sessions or were unable to participate consistently because of injury or illness were excluded from the final analysis. Prior to participation, all players and their legal guardians received detailed information regarding the study procedures, objectives, and potential risks. Written informed consent was obtained from the parents or legal representatives of all participants.

2.3. Intervention Program

The intervention program consisted of an integrated football-specific psychomotor and neuromuscular training model designed for U13 players during the competitive season. The program aimed to support motor coordination, movement efficiency, technical execution, and football-specific functional adaptation under competition conditions.

The intervention was incorporated into the regular training process and implemented throughout the second phase (return round) of the championship season using a structured periodized approach. The training content combined specific physical training with football-specific exercises performed with and without the ball.

Training sessions were conducted 2–3 times per week, with a duration of approximately 90 min per session, according to the competitions calendar and weekly microcycle organization. Psychomotor and neuromuscular activities were primarily incorporated into the introductory and main parts of the training sessions.

The intervention combined coordinative, neuromuscular, technical and tactical components through multidirectional movement tasks and football-specific game situations. The training content included postural stability and postural control exercises, unilateral and bilateral plyometric drills, acceleration and repeated sprint exercises, agility and change-of-direction tasks, coordination ladder and hurdle exercises, and sensorimotor and visual-reaction activities. There were also football-specific technical exercises performed under fatigue conditions, small-sided games, positional play situations, and multidirectional locomotor tasks integrated with passing, shooting, and ball-control exercises.

Several training structures combined explosive muscular actions with immediate football-specific technical execution, including passing accuracy, dribbling, crossing, finishing actions, and transition play. The exercises were progressively adapted according to the players' technical level, functional responses, and competitive demands throughout the intervention period.

The comparative team did not receive the psychomotor intervention and continued its regular football training along the same competitive period.

2.4. Measurements

2.4.1. Psychomotor Performance Assessment

Psychomotor evaluations were performed using standardized motor and neuromuscular performance tests.

Sprint and agility performance were assessed using the Witty wireless electronic timing system (Microgate, Bolzano, Italy). Sprint ability was evaluated using the 20 m sprint test (AV20 m), whereas change-of-direction ability was assessed using the 4 × 5 m shuttle run test (N 4 × 5 m). The analyzed variable was execution time (s) (Table S1, SM1).

Football-specific agility and reactive coordination were additionally evaluated using the agility with ball passing test (APM) performed with the Witty SEM intelligent light system (Microgate, Bolzano, Italy). During the test, participants reacted to visual stimuli while simultaneously performing football passing actions. The analyzed variables included reaction time (s) and total execution time (s) (Table S1, SM1).

Postural stability and postural control were assessed using the Sensamove Mini-Board platform (Sensamove, Utrecht, The Netherlands). Lateral bipedal postural stability (EBL), vertical bipedal postural control (EBV), and vertical unipedal postural control tests for the left and right lower limbs (EUV-LL and EUV-RL) were performed under standardized conditions (Table S1, SM2). The measured outcome was expressed as a percentage (%).

Explosive strength and reactive neuromuscular performance were evaluated using the OptoJump Next system (Microgate, Bolzano, Italy). The testing protocol included Squat Jump (SJ), Countermovement Jump (CMJ), and repeated jumps for 15 s (RJ 15 s) (Table S1, SM3). The analyzed variables included jump height (cm), step frequency (PACE), and reactive strength index (RSI).

All assessments were conducted in the sports hall of CS Otopeni in standardized environmental conditions. Participants performed 2–3 attempts for each test, and the best performance was retained for statistical analysis.

2.4.2. Technical Performance Assessment

Technical performance was evaluated using three football-specific technical tests performed under standardized conditions. The tests were adapted according to the methodological recommendations of the Romanian Football Federation and were designed to assess football-specific technical execution, coordination, precision, and movement efficiency (Table S2, SM4).

Test 1 evaluated shooting accuracy toward different target zones of the goal. Participants performed five shooting attempts directed toward predefined scoring areas with different point values (1, 3, 5, and 10 points), depending on shooting precision and target difficulty.

Test 2 evaluated technical execution frequency and ball-control efficiency under repeated effort conditions. Players performed successive inside-foot ball strikes for 30 s, and the total number of successful repetitions was multiplied by three points to obtain the final score.

Test 3 evaluated long-pass precision. Players executed long aerial passes from a distance of 20 m toward a 5 × 5 m target area marked with cones. Each participant performed three attempts, and each successful execution was awarded 20 points.

The technical assessments analyzed football-specific motor abilities including passing precision, shooting accuracy, execution speed, coordination, and technical efficiency under dynamic conditions. The obtained scores were comparatively analyzed between the beginning and the end of the competitive season to examine changes associated with the intervention period.

2.4.3. Functional Monitoring

Functional monitoring during official football matches was performed using the Polar Vantage V system integrated with Polar Flow software (Polar Electro Oy, Kempele, Finland).

The monitoring system was used to record cardiovascular and locomotor responses while playing competition matches in real ecological conditions. This approach enabled assessing the physiological effort and locomotor demands during an official competition.

The monitored physiological and locomotor variables included (Table S3, SM5): average heart rate (HRavg, bpm), maximal heart rate (HRmax, bpm), average running speed (Savg, km/h), and maximal running speed (Smax, km/h).

Effort intensity distribution was analyzed according to cardiovascular and locomotor intensity zones (Table S3, SM6): Zone 3 (Z3): HR = 141–160 bpm; Speed = 7.01–11.0 km/h; Zone 4 (Z4): HR = 161–180 bpm; Speed = 11.01–15.0 km/h; Zone 5 (Z5): HR = 181–200 bpm; Speed = 15.01–19.0 km/h.

For each intensity zone, relative distribution (%) and exercise duration (min) were recorded and analyzed. The monitoring data provided objective information regarding internal load, locomotor activity, and cardiovascular responses during official matches.

2.4.4. Competitive Performance

Competitive performance indicators were monitored throughout the 2022–2023 championship season in order to analyze performance trends in real competitive conditions during the intervention period.

The competitive analysis involved a comparative assessment between the first phase of the championship season (first round: September–November 2022) and the second phase of the season (second round: March–June 2023), during which the intervention program was implemented. The analyzed competitive indicators included number of wins, draws, losses, goals scored, goals conceded, and overall competitive efficiency during both championship phases.

Official match results from 11 matches in the first round and 11 matches in the second round were analyzed to identify potential competitive performance changes during the intervention period (Table S4, SM7).

Competitive performance data were obtained from the unofficial team performance records of the U13 County Championship (Series 2), 2022–2023 season.

The same technical assessment protocol was applied to the external comparative team at both assessment points to enable contextual comparison of technical performance changes throughout the competition season.

2.5. Statistical Analysis

Statistical analysis was performed using IBM SPSS Statistics software version 23 (IBM Corp., Armonk, NY, USA). Descriptive statistics were calculated for all variables and are presented as mean ± standard deviation (SD). Data distribution normality was assessed by means of the Shapiro–Wilk test. Depending on data distribution, comparative analyses were performed using paired-samples t-tests for normally distributed variables or the Wilcoxon signed-rank test for non-normally distributed variables. Between-group comparisons were performed only for football-specific technical variables because psychomotor and physiological assessments were available exclusively for the intervention group. Between-group comparisons of change scores (Δ = TFA − TIA) were performed using the Mann–Whitney U test.

No formal a priori sample size calculation was performed before data collection because the study included all eligible players available within the participating U13 team (n = 17). A sensitivity analysis indicated that, with n = 17, α = 0.05, and a two-sided paired-samples design, the available sample provides approximately 80% power to detect an effect of approximately d = 0.72. This sensitivity estimate applies to within-participant paired comparisons and should not be interpreted as demonstrating adequate statistical power for correlation analyses or between-group comparisons.

Effect sizes for paired comparisons were quantified using Cohen's d, calculated as the mean individual change divided by the standard deviation of the individual change scores, with change defined as Δ = POST − PRE: d = MΔ/SDΔ. This standardized measure was calculated directly from the paired data and was reported alongside the corresponding inferential test, including the Wilcoxon signed-rank test when data did not meet the assumptions for parametric analysis. The sign of d indicates the direction of change, whereas its absolute magnitude represents the standardized magnitude of the within-participant change. Effect sizes were interpreted as small d = 0.20, moderate d = 0.50, and large d = 0.80.

Relationships between psychomotor, technical, and functional variables were analyzed using Spearman's rank-order correlation coefficients (ϱ) due to the relatively small sample size and the non-normal distribution of several variables. For Spearman's rank-order correlation coefficients (ϱ), 95% confidence intervals (CIs) were calculated using the Fisher z-transformation, with $SE_z = 1/\sqrt{n-3}$, and back-transformed to the correlation scale. No correction for multiple comparisons was applied because the analyses were considered exploratory. Statistical significance was established at $p < 0.05$.

For visualization in the radar chart, relative percentage changes were calculated by means of the measurement-specific baseline. For psychomotor and functional variables assessed before and after the intervention, relative change was calculated as Δ% = [(POST − PRE)/PRE] × 100. For football-specific technical performance, relative change was calculated as Δ% = [(TFA − TIA)/TIA] × 100, using the summed score of the three technical tests (Test 1 + Test 2 + Test 3) at each assessment point. Competitive performance was quantified with the help of match points, calculated as 3 points for a win, 1 point for a draw, and 0 points for a loss, and its relative change was calculated as [(Second round − First

round)/First round] × 100. Negative values indicate a reduction relative to the corresponding baseline. For variables expressed as time, such as sprint performance, a negative percentage change shows a smaller execution time, thus indicating improved performance.

Considering the multidimensional nature of the assessments, Spearman's rank-order correlation analysis was performed to examine the relationships among 28 parameters representing three predefined performance domains: motor, technical, and functional. Because all pairwise relationships among the 28 parameters were examined, a total of 378 Spearman correlation coefficients were calculated. The analysis was used to characterize associations within and across the three domains, with particular emphasis on cross-domain relationships relevant to football-specific motor performance and functional responses.

The correlation analysis focused on the relationships considered most relevant to the study objectives, particularly the associations between variables from different performance domains, including motor, technical, and functional indicators. The complete correlation matrix comprising all 378 pairwise correlations is provided in the Supplementary Materials. No external panel of judges was used for variable selection; the 28 parameters were predefined according to the measurement protocol and their classification within the motor, technical, and functional domains. Given the exploratory nature of the analysis and the absence of adjustment for multiple comparisons, the reported associations were interpreted as exploratory and hypothesis-generating rather than confirmatory.

2.6. Ethical Statement

The study was conducted in accordance with the Declaration of Helsinki and formed part of the broader doctoral research project of the Doctoral School of Sports Science and Physical Education, National University of Science and Technology "Politehnica" Bucharest, University Center Pitești, Romania. The initial institutional ethical approval for the doctoral research was granted by the University of Pitești (Approval No. 7/25.08.2022). Following this approval, the participants and their parents or legal guardians were informed about the planned research procedures, and written informed consent was obtained on 5 September 2022. The subsequent institutional ethical approval dated 12 October 2022 referred specifically to the intervention group within the doctoral research and was not the initial ethical approval for the broader doctoral research project. The implementation of the experimental research was documented by Certificate of Experimental Research Implementation No. 229/12.07.2024. For the present manuscript, retrospective institutional ethical approval for the use and analysis of the previously collected data was granted by the institutional Ethics Committee (Approval No. 37/15.05.2025).

3. Results

3.1. Psychomotor Performance

Significant changes were observed in several psychomotor performance parameters following the intervention period in U13 football players (Table 1, Supplementary Materials, SM1–3).

Table 1. Changes in psychomotor performance parameters following the intervention program in U13 football players (n = 17).

Tests	Parameters	PRE		POST		Paired Diff.		t	(p)	d
		Mean	SD	Mean	SD	Mean	SD			
AV 20 m	Time (sec)	3.98	0.29	3.79	0.25	−0.19	0.17	−4.64	0.0003	−1.13
N 4 × 5 m	Time (sec)	7.83	0.45	7.74	0.57	−0.09	0.43	−0.86	0.401	−0.21
APM	Time (sec)	47.90	2.34	47.27	1.59	−0.62	2.76	−0.93	0.366	−0.22

EBL	Performance (%)	64.53	16.09	84.71	9.47	20.17	14.96	5.56	0.0001	1.35
EBV	Performance (%)	77.00	11.28	86.47	8.56	9.47	12.33	3.17	0.006	0.77
EUV LL	Performance (%)	87.23	8.93	92.47	9.07	5.24	13.63	1.58	0.133	0.38
EUV RL	Performance (%)	84.29	14.75	90.71	10.21	6.41	17.35	1.52	0.147	0.37
CMJ	Height (cm)	22.18	3.78	22.93	3.29	0.75	2.64	1.17	0.260	0.28
J 15 s	Pace (step/s)	1.69	0.08	1.66	0.11	−0.03	0.13	−1.06	0.304	−0.26
	RSI (m/s)	0.77	0.16	0.71	0.18	−0.06	0.14	−1.65	0.118	−0.40

Note: before (PRE) and after (POST) the implementation of the intervention program; Diff.—differences. Effect size interpretation for Cohen's d: 0.20 = small, 0.50 = moderate, 0.80 = large.

Sprint performance over 20 m (AV20 m) improved by −0.19 s (p = 0.0003), with a large effect size (d = −1.13). Lateral bipedal postural stability (EBL) also increased by approximately 20% (p = 0.0001), corresponding to a large effect size (d = 1.35). Vertical bipedal postural control (EBV) improved by 10% (p = 0.006), with a moderate-to-large effect size (d = 0.77). These findings highlight improvements in postural control and neuromotor regulation following the intervention period. No statistically significant differences were identified for agility performance in the 4 × 5 m shuttle run test (N 4 × 5 m), movement anticipation during the agility with ball passing test (APM), vertical unipedal postural control (EUV), countermovement jump performance (CMJ), step frequency (PACE), or reactive strength index (RSI) (p > 0.05).

However, several of these variables had small-to-moderate effect sizes; additionally, these estimates are presented descriptively and should not be interpreted as evidence of adaptation in the absence of statistically significant changes. Overall, the findings indicate that the intervention period was mainly associated with improvements in postural stability, sprint-related motor performance, and psychomotor control in youth football players.

3.2. Technical Performance

Football-specific technical performance improved in both groups during the competition season; however, greater improvements were generally observed in the intervention group following the psychomotor-oriented intervention (Table 2 and Table S2, SM4).

Technical performance was additionally evaluated in an external comparative team that completed the same technical assessments but did not participate in the psychomotor intervention.

Table 2. Within-group changes and between-group comparison of football-specific technical performance following the psychomotor intervention program in U13 football players (n = 17).

Tests		TIA		TFA		Paired Diff.		Z	(*p*)	d	*p* (Δ)
		Mean	SD	Mean	SD	Mean	SD				
Test 1 (points)	EG	11.47	5.05	14.41	5.61	2.94	3.34	3.12	0.002	0.88	0.383
	CG	11.53	3.76	13.65	4.06	2.12	1.36	3.66	0.0003	1.55	
Test 2 (points)	EG	32.65	8.61	34.71	9.24	2.06	2.75	2.44	0.016	0.75	0.186
	CG	32.59	8.93	33.82	8.81	1.24	1.82	2.14	0.035	0.68	
Test 3 (points)	EG	28.24	17.41	34.12	16.98	5.88	9.39	2.24	0.037	0.63	0.544
	CG	29.41	20.15	32.94	19.93	3.53	10.57	1.34	0.233	0.33	

Note: EG—intervention group; CG—comparative group; Technical Initial Assessment (TIA); Technical Final Assessment (TFA); Diff.—differences; p (Δ) indicates the between-group comparison of change scores (intervention group vs. comparative group). Effect size interpretation for d: 0.20 = small, 0.50 = moderate, 0.80 = large.

Within the intervention group, Test 1 (shooting accuracy toward different target zones of the goal) increased by 2.94 points (p = 0.002, d = 0.88), corresponding to a large

standardized change. Test 2 (technical execution frequency and ball-control efficiency under repeated effort conditions) improved by 2.06 points (p = 0.016, d = 0.75), while Test 3 (long-pass precision) increased by 5.88 points (p = 0.037, d = 0.63), both demonstrating moderate standardized change.

The comparative group also showed significant improvements in Test 1 (Δ = 2.12 points, p < 0.001, d = 1.55) and Test 2 (Δ = 1.24 points, p = 0.035, d = 0.68), whereas the change observed in Test 3 did not reach statistical significance (Δ = 3.53 points, p = 0.233, d = 0.33). To further examine the magnitude of change associated with the psychomotor-oriented intervention, the magnitude of change (Δ) in football-specific technical performance was compared between the intervention and comparative groups.

Between-group comparisons of change scores (Δ) did not reveal statistically significant differences for Test 1 (p = 0.383), Test 2 (p = 0.186), or Test 3 (p = 0.544). Although the intervention group showed numerically greater mean improvements across all three technical tests, these differences were not statistically significant. Therefore, the present study does not provide evidence of a detectable advantage of the psychomotor-oriented intervention over regular football training for football-specific technical performance.

Generally, these findings suggest that psychomotor-oriented training may facilitate football-specific technical development during early adolescence. However, because no statistically significant between-group differences were identified, these findings should be interpreted with appropriate methodological caution and confirmed in future controlled studies with larger samples.

3.3. Functional Monitoring

Functional monitoring during official match plays revealed changes in physiological and locomotor variables during the intervention period (Table 3 and Table S3, SM5).

Table 3. Functional physiological responses during match playing before and after the psychomotor intervention in U13 football players (n = 17).

Parameters	PRE		POST		Paired Diff.		Z	(*p*)	d
	Mean	SD	Mean	SD	Mean	SD			
General parameters									
HR avg. (bpm)	131	16.2	131	17.2	0.4	13.8	0.16	0.897	0.03
HR max (bpm)	186	12.2	186	15.5	−0.4	16.1	−0.02	1.000	−0.03
HR min (bpm)	71	6.7	65	11.4	−6.2	13.6	−1.73	0.089	−0.46
Speed avg. (km/h)	2.0	0.8	1.9	0.8	−0.1	0.6	−0.39	0.717	−0.11
Speed max (km/h)	19.3	3.1	18.8	2.8	−0.5	4.9	−0.54	0.602	−0.11
High-intensity zones									
HR Z3 (%)	25	10.8	19	5.9	−5.7	13.7	−1.73	0.089	−0.41
HR Z4 (%)	11	7.43	13	11.7	1.7	17.1	0.21	0.849	0.10
HR Z5 (%)	5	7.82	7	9.4	2.0	14.6	0.36	0.736	0.14
Speed Z3 (min)	1.63	1.09	1.1	0.9	−0.5	0.7	−2.44	0.016	−0.70
Speed Z4 (min)	0.39	0.4	0.21	0.2	−0.2	0.3	−1.75	0.084	−0.56
Speed Z5 (min)	0.1	0.1	0.01	0.02	−0.04	0.1	−1.82	0.074	−0.57

Note: before (PRE) and after (POST) the implementation of the intervention program; d represents Cohen's standardized effect size for paired observations, calculated as the mean of the paired differences divided by the standard deviation of the paired differences. Effect size interpretation for d: 0.20 = small, 0.50 = moderate, 0.80 = large.

No statistically significant changes were identified in general cardiovascular parameters or high-intensity heart rate zones (p > 0.05). Although several non-significant variables showed small-to-moderate standardized effect sizes, these estimates are presented

descriptively and should not be interpreted as evidence of physiological adaptation, particularly given the small sample size and the absence of comparable functional measurements in the comparative group. The only statistically significant functional change was a reduction in Speed Z3 exposure ($p = 0.016$; d = −0.70). Speed Z4 and Speed Z5 exposure also decreased, with moderate standardized effect sizes (d = −0.56 and d = −0.57, respectively), although these changes did not reach statistical significance ($p = 0.084$ and $p = 0.074$). These findings indicate a reduction in the time spent in selected locomotor speed zones during official match play, while the underlying mechanisms and their relationship to the intervention cannot be established from the present longitudinal design.

The observed physiological responses suggest that the intervention period was mainly associated with functional regulation and locomotor organization rather than major changes in overall cardiovascular load during competition.

3.4. Correlation Analysis of Psychomotor, Technical, and Functional Adaptations

Spearman correlation analysis identified significant associations between psychomotor, technical, and functional variables following the intervention period (Table 4).

Table 4. Selected relationships between psychomotor, technical, and functional parameters following the intervention program in U13 football players (n = 17).

Variables	ϱ	95% CI
PACE—TEST2	−0.564	−0.822 to −0.114
EBL—HR min	−0.753	−0.906 to −0.427
EBV—HR min	−0.553	−0.817 to −0.099

Note: Spearman coefficient (ϱ).

A moderate negative relationship was observed between step frequency performance (PACE) and technical execution (TEST2) (ϱ = −0.564), suggesting a possible interaction between neuromuscular rhythm and football-specific motor performance. Strong negative correlations were also identified between postural stability/control performance variables and minimal heart rate responses during match activity. Lateral bipedal postural stability (EBL) demonstrated a strong association with HR min (ϱ = −0.753), while vertical postural control (EBV) was moderately associated with HR min (ϱ = −0.553). These findings show that improvements in postural control may be associated with more efficient physiological regulation during competitive football activity.

Correlation analysis of intervention-related changes (Δ) pointed out several significant associations between psychomotor adaptations and functional responses during competition activity (Table 5).

Table 5. Relationships between intervention-induced changes in psychomotor, technical, and functional parameters in U13 football players (n = 17).

Variables (Δ)	ϱ	95% CI
EURL—TEST2	−0.693	−0.880 to −0.318
N4X5—HR Z3	−0.495	−0.788 to −0.019
EBL—S max	−0.580	−0.829 to −0.138
EBL—S Z5	−0.552	−0.816 to −0.097
EBL—S Z3	−0.488	−0.785 to −0.010
EBV—HR Z3	−0.606	−0.842 to −0.177
CMJ—HR Z3	−0.522	−0.802 to −0.055

Note: Spearman coefficient (ϱ).

A strong negative correlation was identified between vertical unilateral postural control (EUV-RL) and technical execution performance in TEST2 (ϱ = −0.693). Agility-related adaptation (N4 × 5 m) showed a moderate negative association with cardiovascular activity in HR Zone 3 (ϱ = −0.495). Lateral bipedal postural stability (EBL) revealed moderate negative correlations with maximal running speed (S max) (ϱ = −0.580), Speed Zone 5 activity (S Z5) (ϱ = −0.552), and Speed Zone 3 activity (S Z3) (ϱ = −0.488). In addition, vertical bipedal postural control (EBV) was negatively associated with HR Zone 3 responses (ϱ = −0.606). Countermovement jump performance (CMJ) also demonstrated a moderate negative correlation with HR Zone 3 activity (ϱ = −0.522).

All these findings suggest the existence of functional relationships between psychomotor adaptations and physiological responses during competitive football activity.

3.5. Competitive Performance

Competitive performance was examined descriptively by comparing the team's competitive indicators between the first and second rounds of the 2022–2023 championship season. The analysis included match outcomes, goals scored and conceded, and match points recorded before and during the intervention period (Table 6 and Table S4, SM7).

Table 6. Competitive performance indicators recorded during the first and second rounds of the championship season in U13 football players.

Indicators	First Round	Second Round	Δ
Goals scored	14	13	−1
Goals conceded	19	17	−2
Wins	2	3	+1
Draws	1	2	+1
Losses	8	6	−2
Match points	7	11	+4

Note: Δ represents the difference between the results recorded during the second round and those recorded during the first round of the championship season (second round—first round). Positive values indicate an increase, whereas negative values show a decrease in the respective indicator. For goals conceded and number of losses, negative values reveal a reduction in these indicators. Match points were calculated as 3 points for a win, 1 point for a draw, and 0 points for a loss.

The number of wins increased by 50%, whereas draws increased by 100% during the second round. In contrast, match losses decreased by 25%, and goals conceded were reduced by 10.53%. The number of wins increased from 2 to 3, while draws increased from 1 to 2 during the second round. Conversely, the number of losses decreased from 8 to 6, and goals conceded decreased from 19 to 17. Match points increased from 7 in the first round to 11 in the second round, corresponding to a relative increase of 57.14%. Goals scored remained relatively stable, decreasing from 14 to 13 (−7.14%).

These descriptive changes underline an improved competitive record during the second round; however, these findings should be interpreted descriptively rather than as evidence of an intervention effect. This is because competitive performance was not subjected to inferential statistical testing and may also reflect cumulative training, competition exposure, and other contextual factors.

3.6. Multidimensional Functional Profiling and Integrated Correlation Analysis

Figure 2 illustrates the relative percentage changes observed in selected psychomotor, technical, functional, and competitive performance indicators during the intervention period.

The radar profile showed the largest relative changes in lateral bipedal postural control, sprint performance, and competitive efficiency.

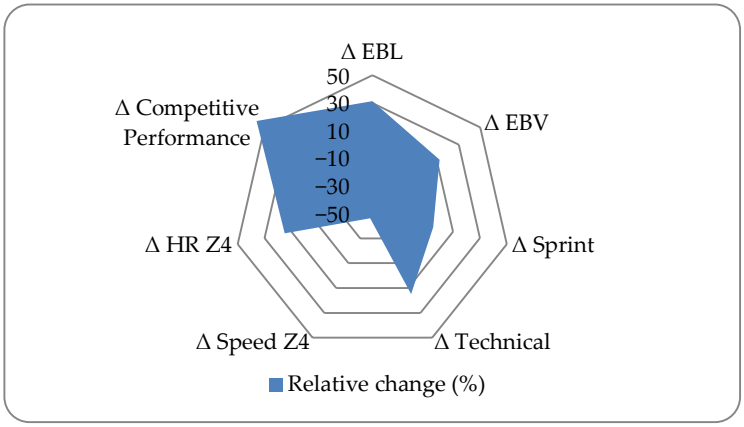


Figure 2. Normalized profile of relative changes in psychomotor, technical, functional, and competitive performance indicators in U13 football players. Psychomotor and functional changes were calculated relative to PRE values; technical change was calculated relative to TIA using the summed score of the three technical tests; competitive performance change was calculated from match points (3 points for a win, 1 point for a draw, and 0 points for a loss). Negative values indicate a reduction relative to the corresponding baseline. For sprint time, a negative change indicates improved performance.

The multidimensional profile showed relative changes across the selected psychomotor, technical, functional, and competitive indicators. The greatest relative changes were observed for competitive performance based on match points (Δ Competitive = 57.14%), locomotor activity in Speed Z4 (Δ Speed Z4 = −46.15%), and lateral bipedal postural stability (Δ EBL = 31.27%). Further relative changes were observed for technical performance (Δ Technical = 15.04%), cardiovascular responses in HR Z4 (Δ HR Z4 = 15.12%), and vertical bipedal postural control (Δ EBV = 12.30%). Sprint performance also improved, with sprint time decreasing by 4.77% from baseline (Δ Sprint = −4.77%).

The integrated radar representation highlights the multidimensional nature of the functional adaptation process during the intervention period. The adaptation profile suggests that the intervention period was mainly associated with improvements in neuromotor regulation, sprint-related motor performance, postural control, and competitive efficiency in youth football players.

Figure 3 illustrates the relationship between changes in lateral bipedal postural stability (ΔEBL, %) and high-intensity locomotor activity (Δ Speed Z5, min) during the intervention period. A moderate negative association was identified between the two variables ($\rho = -0.552$, $p = 0.022$). This association shows that greater increases in EBL were accompanied by greater reductions in Speed Z5 exposure. However, the correlation is associative and does not establish a causal relationship between postural control and locomotor activity.

This relationship may indicate improved movement efficiency and more effective functional regulation during competitive activity. Players demonstrating greater postural adaptation were associated with lower participation in extreme locomotor intensity zones, potentially reflecting more efficient neuromotor organization and economical movement strategies during football-specific performance.

The observed findings are consistent with the hypothesis that psychomotor adaptations may contribute not only to improvements in motor control but also to functional regulation during competitive match activity in youth football players.

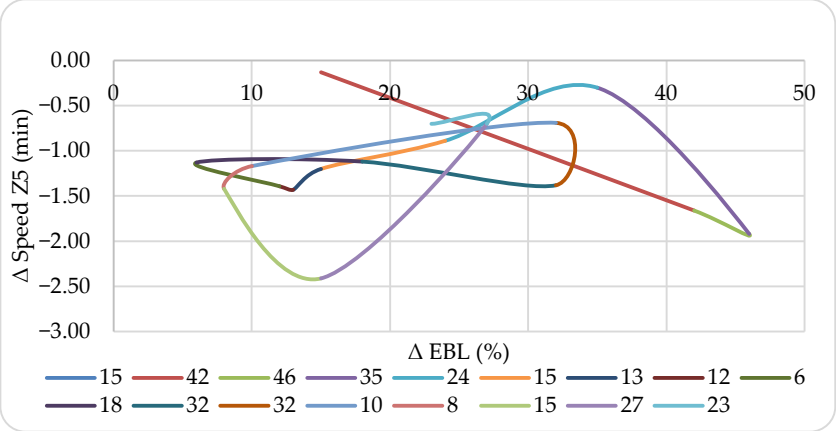


Figure 3. Functional relationship between postural adaptation and high-intensity locomotor performance in youth football players.

4. Discussion

The aim of the present study was to investigate the effects of an integrated psychomotor intervention program on psychomotor abilities, football-specific technical performance, and functional physiological responses in U13 football players during the competition season. The findings are generally consistent with the initial working hypothesis, demonstrating that the psychomotor-oriented training program was associated with improvements in postural control, sprint-related motor performance, football-specific technical execution, and functional competitive adaptation. Moreover, the inclusion of an external comparative group provided additional contextual evidence for interpreting the observed technical adaptations during the competitive season. Significant improvements were identified in sprint performance over 20 m (−0.19 s; $p < 0.001$; d = −1.13), lateral bipedal postural stability (+20%; $p < 0.001$; d = 1.35), and vertical bipedal postural control (+10%; $p = 0.006$; d = 0.77). Furthermore, the relationships identified between psychomotor, technical, and physiological variables imply that neuromotor development may play an important role in performance optimization during early adolescence. Considering the quasi-experimental design and the absence of a randomized control group with complete psychomotor and physiological assessment, the observed findings should be interpreted as intervention-related adaptations rather than definitive causal effects.

One of the most important findings of the present study was the significant improvement observed in bipedal postural stability and control performance (EBL and EBV). These results are consistent with previous research emphasizing the importance of postural control and sensorimotor integration for movement efficiency, technical execution, and functional stability in youth football players [28,29]. During preadolescence, rapid maturation processes within the central nervous system contribute to increased neuroplasticity and motor adaptability, facilitating improvements in postural stability, postural control and coordinative control [30]. The large effect size observed for lateral bipedal postural stability (d = 1.35) indicates a substantial within-group improvement following the intervention period. This finding is consistent with previous research reporting improvements in postural stability and control following integrative neuromuscular training in young soccer players [5]. Furthermore, research comparing neuromuscular and plyometric training in pubertal soccer players has highlighted the multidimensional nature of training-related adaptations, including physical and cognitive outcomes [31]. Although the observed improvement may be compatible with adaptations in neuromuscular control and postural stabilization, these mechanisms were not directly assessed in the present study. Therefore, the findings should be interpreted as evidence of improved postural

stability and control performance rather than as confirmation of specific underlying neuromuscular mechanisms.

Sprint performance over 20 m also improved significantly following the intervention period. Previous studies have shown that sprint ability in youth football is influenced not only by muscular power but also by neuromotor coordination, intermuscular synchronization, and movement organization [32,33]. In this study, the improvement noticed in sprint performance (d = −1.13) may reflect enhanced neuromuscular efficiency associated with the integration of multidirectional coordinative exercises, acceleration tasks, reactive drills, and football-specific movement patterns included in the training process. Considering the developmental characteristics of the U13 age category, these adaptations may also reflect the sensitivity of early adolescence to coordinative and neuromotor stimuli [34].

An important aspect of the findings is that the intervention effects were not generalized across all assessed variables, suggesting selective adaptation patterns depending on the functional specificity of the psychomotor stimuli. Significant improvements were mainly identified in postural stability and control, sprint performance, and technical execution, whereas variables such as CMJ, RSI, unilateral postural control, and several physiological indicators demonstrated only small-to-moderate changes. These findings may indicate that the intervention period predominantly stimulated coordinative, postural, and sensorimotor mechanisms rather than maximal explosive neuromuscular output. In addition, the relatively short duration of the intervention period and the competitive-season training context may have limited the magnitude of strength- and power-related adaptations [35,36].

Significant within-group improvements were observed in all three football-specific technical tests in the intervention group, while the comparative (control) group also improved in two of the three tests. However, between-group comparisons of change scores did not reveal statistically significant differences for any of the three technical tests. Therefore, the present findings do not demonstrate a detectable between-group difference in technical performance associated with the psychomotor-oriented intervention. The observed within-group improvements should be interpreted in the context of the quasi-experimental design and the longer TIA–TFA assessment interval, which included cumulative training and competitive exposure throughout the season. Previous research has shown that balance control, coordination abilities, and sensorimotor integration contribute directly to football-specific motor execution and technical precision in youth athletes [37,38]. The improvements identified in technical execution may be associated with changes in movement organization and perceptual-motor regulation during football-specific actions. These findings provide a rationale for considering the integration of coordinative and neuromotor exercises within technical training structures during sensitive developmental stages [39–41].

Functional monitoring during official match play identified changes in physiological and locomotor variables during the intervention period. No statistically significant changes were identified in general cardiovascular parameters or high-intensity heart rate zones ($p > 0.05$). The only statistically significant change was a reduction in Speed Z3 exposure ($p = 0.016$; d = −0.70). Speed Z4 and Speed Z5 exposure also decreased, with moderate standardized effect sizes (d = −0.56 and d = −0.57, respectively), although these changes were not statistically significant. These findings indicate changes in locomotor exposure during official match play; however, the underlying mechanisms cannot be established from the present longitudinal design, particularly given the absence of comparable functional measurements in the comparative group and potential differences in playing-time exposure. Nevertheless, the intervention group showed numerically greater mean improvements across all three technical tests, although these differences were not

statistically significant between groups. This pattern may represent a potentially relevant practical observation, but it cannot be attributed specifically to the psychomotor-oriented intervention. Similar tendencies have been reported in previous studies investigating physiological adaptation and locomotor efficiency in youth football players, where improved coordination and neuromuscular regulation have been associated with more economical movement patterns and reduced unnecessary high-intensity activity during match play [42,43]. Furthermore, contemporary locomotor profiling studies have emphasized the importance of integrated monitoring approaches for understanding adaptation processes during youth football development [19,20].

The correlation analysis additionally highlighted the integrated nature of psychomotor adaptation in youth football. Significant associations identified between balance performance, sprint-related variables, technical execution, and physiological responses reveal complex interactions between neuromotor regulation and functional adaptation during competitive activity. In particular, negative relationships were observed between postural stability and control variables and high-intensity locomotor zones ($\varrho = -0.488$ to -0.693; $p < 0.05$). This may indicate an association between postural control, movement organization, and physiological regulation during football match play. These findings are partially supported by previous studies investigating interactions between postural regulation, locomotor efficiency, and neuromuscular control in youth athletes [24,29]. However, these correlational findings should be interpreted cautiously due to the relatively small sample size and the exploratory nature of the multidimensional analysis.

The analysis of intervention-related changes (Δ) further supports the multidimensional character of adaptation processes in youth football. Relationships identified between sprint improvement, heart rate responses, and locomotor activity suggest that neuromotor adaptations may influence physiological demands during competitive performance [20,31]. Likewise, the associations between balance-related adaptations and locomotor intensity zones may reflect improved functional regulation and coordinative efficiency during football-specific activity [17,36]. These findings are consistent with contemporary perspectives emphasizing the interaction between neuromuscular regulation, locomotor organization, and sport-specific motor performance during adolescence [34].

An additional contribution of this study is represented by the radar-based functional profiling approach used to visualize multidimensional adaptation patterns. The normalized profile revealed the largest adaptations for lateral bipedal postural stability (ΔEBL = 31.27%), competitive performance (ΔCompetitive = 57.14%), and sprint-related motor performance (ΔSprint = −4.77%). Although the radar model should be considered primarily as a practical descriptive tool rather than a validated diagnostic method, it facilitated the identification of individualized adaptation tendencies across psychomotor, technical, physiological, and competitive domains. From an applied perspective, this type of integrated visualization may support coaches in individualized training management and functional monitoring during long-term athlete development [23,24].

Match-related locomotor and physiological indicators may have been influenced by differences in playing time, player role, tactical demands, and match context between monitoring sessions. As playing-time exposure was not standardized or normalized across all players and assessment periods, changes in locomotor intensity zones should be interpreted cautiously and cannot be attributed exclusively to changes in movement efficiency or physiological adaptation.

The originality of the present study resides in the integrated analysis of psychomotor, technical, physiological, and competitive indicators within a real ecological football environment during preadolescence. Unlike many previous investigations focusing on isolated performance components, the current study attempted to analyze the multidimen-

sional interactions between neuromotor adaptation, technical execution, locomotor responses, and competitive efficiency during the competitions. An additional strength of the present investigation is the inclusion of an external comparative team for football-specific technical assessment, providing a contextual benchmark for interpreting intervention-related technical adaptations under real competitive conditions.

From an applied perspective, the present findings highlight the potential relevance of integrating psychomotor-oriented exercises, including balance, coordination, reactive agility, and sensorimotor activities, into youth football training. However, although significant within-group improvements were observed in several psychomotor and technical measures, between-group comparisons of change scores did not reveal statistically significant differences in football-specific technical performance. Therefore, the observed technical improvements cannot be attributed specifically to the psychomotor-oriented intervention and may also reflect cumulative training, competition exposure, and other contextual factors during the season. The findings should consequently be interpreted as preliminary and hypothesis-generating. Future controlled studies with larger samples, standardized training exposure, and complete multidimensional assessment are warranted to further examine the potential contribution of psychomotor-oriented training to technical and functional development during early adolescence.

Limitations and Future Directions

Biological maturation was not assessed using maturity-offset or peak-height-velocity methods and therefore could not be included as a covariate or used to stratify participants. Because growth and maturational status may substantially influence sprint, balance, strength, and technical performance during early adolescence, maturation represents an important potential confounder of the observed longitudinal changes.

Several limitations of this study should be acknowledged. First, the relatively small sample size and the inclusion of players from a single football team limit the generalizability of the findings to broader youth football populations. The limited number of participants was primarily associated with the restricted availability of Polar monitoring devices and psychomotor assessment equipment, which limited the feasibility of implementing a parallel comparison group under equivalent multidimensional assessment conditions.

Second, although the study did not include a randomized control group undergoing complete multidimensional assessment, an external comparative team was evaluated exclusively for football-specific technical performance at both assessment moments. This comparison provided a contextual reference for interpreting the observed technical adaptations. However, causal inference remains limited because the comparative analyses were restricted to technical variables and were not extended to psychomotor or physiological assessments. Consequently, the findings should be interpreted as intervention-related adaptations identified under real competition conditions rather than definitive evidence of intervention superiority.

Another limitation concerns the selective inclusion of representative psychomotor, technical, functional, and competitive variables in the final statistical and graphical analyses. Although this approach improved the interpretability of the multidimensional data and reduced analytical complexity, some potentially relevant performance indicators were not included in the final correlational models and graphical representations.

Furthermore, the intervention implementation and operational training management were conducted by a single coach-researcher, which ensured methodological consistency throughout the competitive season but may also represent a potential source of observational bias. Functional monitoring during official matches was additionally performed

only during selected competition periods due to the limited availability of the Polar monitoring system, making continuous longitudinal monitoring across the entire season unfeasible.

The technical assessments were performed at the beginning and end of the competition season, whereas the psychomotor intervention was implemented only during the final 12 weeks. Therefore, changes in football-specific technical performance between TIA and TFA may also reflect cumulative training and competition exposure during the earlier part of the season and cannot be attributed exclusively to the intervention.

The long-pass precision test was based on three attempts per player. Although this procedure was consistent across assessments, the limited number of trials may increase measurement variability and should be considered when interpreting changes in this technical outcome.

Despite these limitations, the study provides practically relevant information regarding the multidimensional effects of psychomotor-oriented interventions in youth football players under real competition conditions. The integrated analysis of psychomotor, technical, physiological, and competitive indicators represents an applied perspective for understanding functional adaptation processes during preadolescence.

Future research should investigate larger groups of youth football players using controlled longitudinal designs and multidimensional functional profiling approaches integrating neuromotor, biomechanical, physiological, and technical–tactical assessment systems. Further studies should also examine the long-term transfer effects of psychomotor-oriented interventions on tactical behavior, movement economy, injury prevention, and competitive performance development throughout adolescence.

5. Conclusions

The 12-week integrated psychomotor-oriented intervention was associated with significant within-group improvements in 20-m sprint performance and bipedal postural stability and control, with large standardized effects for sprint performance and lateral bipedal postural stability. Significant within-group improvements were also observed in the three football-specific technical tests; however, between-group comparisons of change scores did not reveal statistically significant differences between the intervention and comparative groups.

Functional monitoring showed a significant reduction in Speed Z3 exposure, whereas the remaining physiological and locomotor variables did not show statistically significant changes. The observed correlations among motor, technical, and functional variables highlight potentially relevant cross-domain associations, but these findings are exploratory and do not establish causality.

Given the small sample, non-randomized design, incomplete multidimensional assessment in the comparative group, longer technical assessment interval, and absence of biological maturation assessment, the findings should be considered preliminary evidence of within-group changes associated with psychomotor-oriented training during preadolescence.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/app162010092/s1. Individual raw psychomotor, technical, and functional assessment data; intervention-induced changes (Δ values and relative percentage changes) for all analyzed variables; complete statistical outputs including Shapiro–Wilk normality tests, paired comparative analyses, effect sizes, and exact *p*-values; complete Spearman correlation matrices for post-intervention variables and intervention-induced changes (Δ); graphical representation of the psychomotor intervention structure and multidimensional functional profiling model.

Author Contributions: V.P. and B.A.P.; methodology, V.P., B.A.P., I.M. (Ilie Mihai) and P.F.V.; software, V.P.; validation, L.E.M., I.M. (Ion Mihaila) and C.C.; formal analysis, V.P. and B.A.P.; investigation, V.P., B.A.P., I.M. (Ilie Mihai) and P.F.V.; resources, B.A.P., I.M. (Ilie Mihai) and P.F.V.; data curation, V.P., B.A.P. and I.M. (Ilie Mihai); writing—original draft, V.P. and B.A.P.; writing—review and editing, L.E.M., I.M. (Ion Mihaila), N.L.M., C.C. and D.B.; visualization, V.P.; supervision, L.E.M., I.M. (Ion Mihaila), N.L.M. and D.B.; project administration, V.P.; funding acquisition, V.P. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki and formed part of the broader doctoral research project of the Doctoral School of Sports Science and Physical Education, National University of Science and Technology "Politehnica" Bucharest, University Center Pitești, Romania. The initial institutional ethical approval for the doctoral research was granted by the University of Pitești (Approval No. 7/25.08.2022). The subsequent institutional ethical approval dated 12 October 2022 referred specifically to the intervention group within the doctoral research. The implementation of the experimental research was documented by Certificate of Experimental Research Implementation No. 229/12.07.2024. For the present manuscript, retrospective ethical approval for the use and analysis of the previously collected data was granted by the institutional Ethics Committee (Approval No. 37/15.05.2025).

Informed Consent Statement: Following the initial ethical approval, the participants and their parents or legal guardians were informed about the purpose and procedures of the planned research. Written informed consent was obtained on 5 September 2022 from the parents or legal guardians of all participants, who were minors at the time of the study.

Data Availability Statement: The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request. The original contributions presented in this study are included in the article/Supplementary Material. Further inquiries can be directed to the corresponding author. Data are not publicly available due to privacy and ethical restrictions involving minor participants.

Acknowledgments: This study was conducted within the doctoral research framework of the Doctoral School of Science of Sports and Physical Education, National University of Science and Technology POLITEHNICA Bucharest, University Center of Pitești, Romania. The authors gratefully acknowledge the Human Performance Research Center of the Department of Physical Education and Sport for providing access to the research infrastructure and assessment equipment used in this study. The authors also thank CS Otopeni, Romania, for facilitating the implementation of the intervention program and the competitive monitoring procedures. During the preparation of this manuscript, generative artificial intelligence tools were used to support language refinement and academic editing. All generated content was critically reviewed, revised, and validated by the authors, who take full responsibility for the accuracy, integrity, and scientific validity of the published work.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

APM	agility with ball passing test
AV 20 m	sprint test 20 m
avg	Average (mean)
CI	confidence interval
CMJ	Countermovement Jump

Cohen's d	standardized effect size for paired observations.
CS	Sports Club
Diff.	Differences
dev	deviation
EBL	bipedal lateral postural stability
EBV	bipedal vertical postural control
EUV-LL	left lower limb
EUV-RL	right lower limb
HRavg	average heart rate
HRmax	maximal heart rate
HRmin	minimal heart rate
N 4 × 5 m	4 × 5 m shuttle run test
PACE	step frequency
Perf.	performance
RSI	Reactive Strength Index
Savg	average running speed
SD	Standard deviation
SJ	Squat Jump
SM	Supplementary Materials
Smax	maximal running speed
t	Parametric paired-samples t-tests for normally distributed variables
TContact	contact time
TFlight	flight time
TFA	Technical Final Assessment
TIA	Technical Initial Assessment
W	Body Weight
z	Nonparametric Wilcoxon signed-rank test for non-normally distributed variables
ϱ	Spearman coefficient

References

1. Vicol, S.I.; Teodorescu, S.; Roșca, R.; Predoiu, R. Psychomotor and cognitive abilities in U14 football players: A comparative analysis across playing positions. *Discobolul—Phys. Educ. Sport Kinetother. J.* **2025**, *64*, 27–40. https://doi.org/10.35189/dpeskj.2025.64.1.3.
2. Leşco, V.; Platonov, V.; Dorgan, V.; Branişte, G. Correlation of sports training with biological and psychomotor development in children and juniors. *Rev. Rom. Educ. Multidimens.* **2025**, *17*, 819–848. https://doi.org/10.18662/rrem/17.4/1080.
3. Boncuț, M.; Mancini, N.; Hăisan, A.-A.; Boncuț, D.; Grosu, E.F.; Popovici, C.; Hervás-Gómez, C.; Man, C.M.; Mancini, S.; Guerriero, M.A.; et al. Technology-Assisted Neuromotor Training for Improving Visuomotor Reaction Performance, Change-of-Direction Quickness, and Bilateral Task Execution in Junior Handball Players. *J. Funct. Morphol. Kinesiol.* **2026**, *11*, 42. https://doi.org/10.3390/jfmk11010042.
4. Zwierko, T.; Nowakowska, A.; Jedziniak, W.; Popowczak, M.; Domaradzki, J.; Kubaszewska, J.; Kaczmarczyk, M.; Ciechanowicz, A. Contributing factors to sensorimotor adaptability in reactive agility performance in youth athletes. *J. Hum. Kinet.* **2022**, *83*, 39–48. https://doi.org/10.2478/hukin-2022-0067.
5. Menezes, G.B.; Alexandre, D.R.O.; Pinto, J.C.B.L.; Assis, T.V.L.; Faigenbaum, A.D.; Mortatti, A.L. Effects of integrative neuromuscular training on motor performance in prepubertal soccer players. *J. Strength Cond. Res.* **2022**, *36*, 1667–1674.
6. Zemková, E. Cognition and Motion: Sensory Processing and Motor Skill Performance in Athletic Training and Rehabilitation. *Appl. Sci.* **2022**, *12*, 10345. https://doi.org/10.3390/app122010345.
7. Lola, A.; Bassa, E.; Symeonidou, S.; Stavropoulou, G.; Papavasileiou, A.; Fregidis, K.; Bismpos, M. Analysis of Motor and Perceptual–Cognitive Performance in Young Soccer Players: Insights into Training Experience and Biological Maturation. *Sports* **2026**, *14*, 22. https://doi.org/10.3390/sports14010022.
8. Keerthika, N.; Kiruthika, V. Integrating electroencephalography with motophysic training: Assessing the shift in fitness and cognitive functions in elite youth soccer players. *Heliyon* **2024**, *10*, e34342. https://doi.org/10.1016/j.heliyon.2024.e34342.

9. Msaidie, H.; Gomez-Merino, D.; Chennaoui, M. Applying sequences of combinations of movements based on motor development stages principles to elite athletes: Conceptual foundations and testable hypotheses. *Front. Sports Act. Living* **2026**, *8*, 1755089. https://doi.org/10.3389/fspor.2026.1755089.
10. Silva-Moya, G.; Méndez-Rebolledo, G.; Valdes-Badilla, P.; Gómez-Álvarez, N.; Guzmán-Muñoz, E. Effects of neuromuscular training on psychomotor development and active joint position sense in school children. *J. Mot. Behav.* **2022**, *54*, 57–66. https://doi.org/10.1080/00222895.2021.1887072.
11. Camenidis, C.M.; de Hillerin, P.J.; Mihăilescu, L.; Geantă, V.A.; Roşu, D. The effects of integrated proprioception development strategies in increasing the psycho-neuro-motor level of children. *Rev. Romaneasca Pentru Educ. Multidimens.* **2024**, *16*, 277–308. https://doi.org/10.18662/rrem/16.3/894.
12. Fard, R.B.; Atri, A.E.; Yazdi, N.K.K.; Shahtahmassebi, B. Assessing changes in static and dynamic postural stability in youth football players following the FIFA 11+ injury prevention program. *Sci. Sports* **2022**, *32*, 215.e1–215.e8. https://doi.org/10.1016/j.scispo.2021.07.003.
13. Mitrousis, I.; Bourdas, D.I.; Kounalakis, S.; Bekris, E.; Mitrotasios, M.; Kostopoulos, N.; Ktistakis, I.E.; Zacharakis, E. The effect of a balance training program on the balance and technical skills of adolescent soccer players. *J. Sports Sci. Med.* **2023**, *22*, 645–657. https://doi.org/10.52082/jssm.2023.645.
14. Haddad, M. Motor Asymmetry in Football: Implications for Muscular Power, Balance, and Injury Prevention. *Symmetry* **2024**, *16*, 1485. https://doi.org/10.3390/sym16111485.
15. Kluczyński, S.; Korzan, K.; Sorek, P.; Jurys, T.; Knapik, A.; Brzęk, A. The Role of Stabilization Exercise in Preventing Pain and Postural Defects in Young Football Players. *Healthcare* **2025**, *13*, 1983. https://doi.org/10.3390/healthcare13161983.
16. Zouhal, H.; Abderrahman, A.B.; Dupont, G.; Truptin, P.; Le Bris, R.; Le Postec, E.; Sghaeir, Z.; Brughelli, M.; Granacher, U.; Bideau, B. Effects of Neuromuscular Training on Agility Performance in Elite Soccer Players. *Front Physiol.* **2019**, *23*, 947. https://doi.org/10.3389/fphys.2019.00947.
17. Menezes, G.B.; Oliveira, R.S.; Ferreira, A.B.; Assis, T.V.; Batista, E.S.; Oliver, J.L.; Lloyd, R.S.; Mortatti, A.L. Does motor coordination influence perceptual-cognitive and physical factors of agility in young soccer players in a sport-specific agility task? *Sports Biomech.* **2024**, *23*, 2108–2121. https://doi.org/10.1080/14763141.2021.1995476.
18. Paśko, W.; Śliż, M.; Paszkowski, M.; Zieliński, J.; Polak, K.; Huzarski, M.; Przednowek, K. Characteristics of Cognitive Abilities among Youths Practicing Football. *Int. J. Environ. Res. Public Health* **2021**, *18*, 1371. https://doi.org/10.3390/ijerph18041371.
19. Clemente, F.M.; Silva, A.F.; Kawczyński, A.; Yıldız, M.; Chen, Y.S.; Birlik, S.; Nobari, H.; Akyildiz, Z. Physiological and Locomotor Demands during Small-Sided Games Are Related to Match Demands and Physical Fitness? A Study Conducted on Youth Soccer Players. *BMC Sports Sci. Med. Rehabil.* **2022**, *14*, 138. https://doi.org/10.1186/s13102-022-00535-w.
20. Silva, A.F.; Oliveira, R.; Ceylan, H.I.; Akyildiz, Z.; González-Fernández, F.T.; Nobari, H.; Yildiz, M.; Birlik, S.; Clemente, F.M. Effects of a Small-Sided Games Training Program in Youth Male Soccer Players: Variations of the Locomotor Profile while Interacting with Baseline Level and Workload. *BMC Sports Sci. Med. Rehabil.* **2022**, *14*, 122. https://doi.org/10.1186/s13102-022-00595-y.
21. Teixeira, J.E.; Afonso, P.; Schneider, A.; Branquinho, L.; Maio, E.; Ferraz, R.; Nascimento, R.; Morgans, R.; Barbosa, T.M.; Monteiro, A.M.; et al. Player Tracking Data and Psychophysiological Features Associated with Mental Fatigue in U15, U17, and U19 Male Football Players: A Machine Learning Approach. *Appl. Sci.* **2025**, *15*, 3718. https://doi.org/10.3390/app15073718.
22. Afonso, P.; Forte, P.; Branquinho, L.; Ferraz, R.; Garrido, N.D.; Teixeira, J.E. Artificial Intelligence in Sub-Elite Youth Football Players: Predicting Recovery Through Machine Learning Integration of Physical, Technical, Tactical and Maturational Data. *Healthcare* **2025**, *13*, 3301. https://doi.org/10.3390/healthcare13243301.
23. Golubev, D.V.; Shchedrina, Y.A.; Melnikov, D.S. Modeling Adaptive Reactions of Young 15–17-Year-Old Professional Football Players' Functional State Based on Integral Monitoring. *Russ. J. Phys. Educ. Sport* **2020**, *15*, 142–151. https://doi.org/10.14526/2070-4798-2020-15-1-142-151.
24. Crivelli, D.; Balconi, M. Neuroassessment in Sports: An Integrative Approach for Performance and Potential Evaluation in Athletes. *Front. Psychol.* **2022**, *13*, 747852. https://doi.org/10.3389/fpsyg.2022.747852.
25. Wilczyński, B.; Biały, M.; Zorena, K. An Interpretable Machine Learning Framework for Athlete Motor Profiling Using Multi-Domain Field Assessments: A Proof-of-Concept Study. *Appl. Sci.* **2025**, *15*, 6436. https://doi.org/10.3390/app15126436.
26. Scano, A.; Guanziroli, E.; Brambilla, C.; Amendola, C.; Pirovano, I.; Gasperini, G.; Molteni, F.; Spinelli, L.; Molinari Tosatti, L.; Rizzo, G.; et al. A Narrative Review on Multi-Domain Instrumental Approaches to Evaluate Neuromotor Function in Rehabilitation. *Healthcare* **2023**, *11*, 2282. https://doi.org/10.3390/healthcare11162282.

27. Gnanasigamani, M.; Kaczmarek, A.; Nosal, J.; Murawska-Ciałowicz, E.; Chmura, P. Multidomain Connectivity as a Marker of HIIT-Induced Adaptation in Elite Youth Soccer Players: A Correlational Mapping Approach. *Appl. Sci.* **2025**, *15*, 7550. https://doi.org/10.3390/app15137550.
28. Hrysomallis, C. Balance Ability and Athletic Performance. *Sports Med.* **2011**, *41*, 221–232. https://doi.org/10.2165/11538560-000000000-00000.
29. Paillard, T. Plasticity of the Postural Function to Sport and/or Motor Experience. *Neurosci. Biobehav. Rev.* **2017**, *72*, 129–152. https://doi.org/10.1016/j.neubiorev.2016.11.015.
30. Myer, G.D.; Faigenbaum, A.D.; Ford, K.R.; Best, T.M.; Bergeron, M.F.; Hewett, T.E. When to Initiate Integrative Neuromuscular Training to Reduce Sports-Related Injuries and Enhance Health in Youth? *Curr. Sports Med. Rep.* **2011**, *10*, 155–166. https://doi.org/10.1249/JSR.0b013e31821b1442.
31. Hammami, A.; Mahmoudi, A.; Selmi, W.; Negra, Y.; Rebai, H.; Granacher, U.; Hammami, R. Effects of neuromuscular versus plyometric training on physical fitness and mental well-being in male pubertal soccer players. *Sci. Rep.* **2025**, *15*, 43393. https://doi.org/10.1038/s41598-025-30142-x.
32. Meylan, C.; Malatesta, D. Effects of In-Season Plyometric Training within Soccer Practice on Explosive Actions of Young Players. *J. Strength Cond. Res.* **2009**, *23*, 2605–2613. https://doi.org/10.1519/JSC.0b013e3181b1f330.
33. Faude, O.; Koch, T.; Meyer, T. Straight Sprinting Is the Most Frequent Action in Goal Situations in Professional Football. *J. Sports Sci.* **2012**, *30*, 625–631. https://doi.org/10.1080/02640414.2012.665940.
34. Synott, G.; Byrne, P.; Lisa, R.; Moody, J.; Duggan, J. The Effects of an In-Season Athletic Motor Skill Competencies Intervention on Physical Bio-Motor Qualities among Elite Youth Soccer Players. *Int. J. Strength Cond.* **2025**, *5*, 1. https://doi.org/10.47206/ijsc.v5i1.475.
35. Silva, A.F.; Alvurdu, S.; Akyildiz, Z.; Badicu, G.; Greco, G.; Clemente, F.M. Variations of the Locomotor Profile, Sprinting, Change-of-Direction, and Jumping Performances in Youth Soccer Players: Interactions between Playing Positions and Age-Groups. *Int. J. Environ. Res. Public Health* **2022**, *19*, 998. https://doi.org/10.3390/ijerph19020998.
36. Hermassi, S.; Konukman, F.; Al-Marri, S.S.; Hayes, L.D.; Bartels, T.; Schwesig, R. Associations between Biological Maturation, Physical Performance, Postural Control, and Mathematical Achievement in Youth Soccer Players. *PLoS ONE* **2024**, *19*, e0298301. https://doi.org/10.1371/journal.pone.0298301.
37. Vestberg, T.; Gustafson, R.; Maurex, L.; Ingvar, M.; Petrovic, P. Executive Functions Predict the Success of Top-Soccer Players. *PLoS ONE* **2012**, *7*, e34731. https://doi.org/10.1371/journal.pone.0034731.
38. Forsman, H.; Blomqvist, M.; Davids, K.; Liukkonen, J.; Konttinen, N. Identifying Technical, Physiological, Tactical and Psychological Characteristics That Contribute to Career Progression in Soccer. *Int. J. Sports Sci. Coach.* **2016**, *11*, 505–513. https://doi.org/10.1177/1747954116655051.
39. Szwarc, A.; Dolański, B.; Wasielewski, K. Relationship between Psychomotor Skills and Game Performance in Youth Soccer Players. *J. Phys. Educ. Sport* **2021**, *21*, 1744–1750. https://doi.org/10.7752/jpes.2021.04221.
40. Mohammed, M.M.; Latif, A.L.M.A. Psychomotor Exercises and Their Impact on Enhancing the Skill Performance of Football Players at the Espanyol Academy Aged 16–18 Years. *Mod. Sport* **2026**, *24*, 5. https://doi.org/10.54702/2708-3454.2108.
41. Abate Daga, F.; Sannicandro, I.; Tanturli, A.; Agostino, S. Youth Soccer Development After a Forced Training Interruption: A Retrospective Analysis of Prepubertal Players. *Sports* **2025**, *13*, 435. https://doi.org/10.3390/sports13120435.
42. Hill-Haas, S.V.; Dawson, B.; Impellizzeri, F.M.; Coutts, A.J. Physiology of Small-Sided Games Training in Football: A Systematic Review. *Sports Med.* **2011**, *41*, 199–220. https://doi.org/10.2165/11539740-000000000-00000.
43. Buchheit, M.; Laursen, P.B. High-Intensity Interval Training, Solutions to the Programming Puzzle. *Sports Med.* **2013**, *43*, 313–338. https://doi.org/10.1007/s40279-013-0029-x.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.